Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **automation-heavy backend workflows**, **Python/Java services**, and AI-enabled systems, with experience in CI/CD, Linux-style tooling, APIs, reliability analysis, and performance-focused debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Restructured a fault-tolerant asynchronous LLM orchestration system using FastAPI background workers, processing 1K+ AI jobs with structured state transitions and improving throughput by **3x** under concurrent workloads.
- Built ETL-style integration pipelines using FastAPI, Supabase, SQL Server, and cloud storage, processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases for **99%+** reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed a concurrent batch-processing pipeline using multithreading and secure file ingestion for PDF, DOCX, and ZIP workflows, processing 1K+ files per batch with fault-tolerant execution.
- Implemented production backend services on AWS serving 150+ instructors and 10,000+ grading requests daily, reducing feedback turnaround by **60%** while maintaining stable API response under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible evaluation and system-behavior analysis while improving debugging quality and benchmark consistency for model-driven workloads.
- Produced technical reports and targeted validation scenarios translating findings into actionable improvements, strengthening cross-team communication, reducing handoff ambiguity, and supporting better engineering decisions across collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable data movement, structured delivery, and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Automated deployments using Jenkins CI/CD pipelines with Docker-based packaging and staging promotion, cutting release time by **50%** and improving deployment consistency across iterative delivery cycles.
- Deployed Azure SQL with automated testing and monitoring workflows, achieving 95%+ successful build validation while improving production-readiness and visibility into release quality across environments.

Skills

Languages: **Python**, **Java**, JavaScript/TypeScript, SQL, Bash, C++
Automation & Testing: automated test suites, CI/CD, Jenkins, PyTest, Mockito, validation and reconciliation workflows
Platform & APIs: FastAPI, Flask, REST APIs, microservices, Linux, SQL Server, PostgreSQL, cloud storage, system integration
Cloud & AI: AWS, Azure, Docker, Kubernetes, LLM applications, RAG concepts, performance debugging, reliability analysis

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document platform with semantic search, summarization, and collaborative Q&A, demonstrating RAG-style workflows over 50K+ documents while reducing document-analysis time by **60%**.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized RESTful data processing API on AWS with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production data workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, maintaining Java and TypeScript modules that accelerated schema exploration, query authoring, and structured debugging of data workflows.